Dictionary

1. Создаем источник для словаря
2. **create table** user_email_source (user_id UInt64, email String) **primary key** user_id

```
CREATE TABLE user_email_source
(
    `user_id` UInt64,
    `email` String
)
PRIMARY KEY user_id

Query id: 83396171-b092-41a8-901d-54f92daa2ecc

Ok.

0 rows in set. Elapsed: 0.063 sec.
```

3. Загружаем данные из файла

```
 clickhouse-client -q "INSERT INTO user_email_source FORMAT CSV" < user_email.csv
yc-user@clickhouse-node:~$ clickhouse-client
ClickHouse client version 25.3.5.42 (official build).
Connecting to localhost:9000 as user default.
Connected to ClickHouse server version 25.3.5.

Warnings:
 * Delay accounting is not enabled, OSIOWaitMicroseconds will not be gathered. You can enable it
using `echo 1 > /proc/sys/kernel/task_delayacct` or by using sysctl.

clickhouse-node.ru-central1.internal :) select * from user_email_source

SELECT *
FROM user_email_source

Query id: 52d6ce6c-87c5-46f2-ad6a-9530fe2647e1
```

	user_id	email
1.	1	as@as.ru
2.	2	ad@ad.ru
3.	3	asa@as.ru
4.	4	ada@ad.ru

```
4 rows in set. Elapsed: 0.002 sec.
```

4. Создаем сам словарь

```
CREATE DICTIONARY user_email
(
    user_id UInt64,
    email String
)
```

```
PRIMARY KEY user_id
SOURCE(CLICKHOUSE(TABLE 'user_email_source'))
LAYOUT(FLAT())
LIFETIME(MIN 0 MAX 1000)


CREATE DICTIONARY user_email
(
    `user_id` UInt64,
    `email` String
)
PRIMARY KEY user_id
SOURCE(CLICKHOUSE(TABLE 'user_email_source'))
LIFETIME(MIN 0 MAX 1000)
LAYOUT(FLAT())

Query id: de1b1f9a-38e1-44e5-82b8-4c19aa443a43

Ok.

0 rows in set. Elapsed: 0.037 sec.
```

5. Создаем таблицу с данными

```
create table user_action (user_id UInt64, action String, expense UInt64) Primary key user_id

CREATE TABLE user_action
(
    `user_id` UInt64,
    `action` String,
    `expense` UInt64
)
PRIMARY KEY user_id

Query id: 071922ec-5a48-435d-9b1a-da00150238da

Ok.

0 rows in set. Elapsed: 0.028 sec.
```

6. Заполняем ее

```
insert into user_action  values(1, 'store', 2),(1, 'drop', 3), (2,'store',2)

INSERT INTO user_action FORMAT Values

Query id: 90238b87-24ca-4328-a2a4-928c088c11cb

Ok.

3 rows in set. Elapsed: 0.011 sec.

clickhouse-node.ru-central1.internal :) select * from user_action

SELECT *
FROM user_action

Query id: dd0c5083-e5e0-4394-9500-062bab3f9f92
```

	user_id	action	expense
1.	1	store	2
2.	1	drop	3
3.	2	store	2

```
3 rows in set. Elapsed: 0.004 sec.
```

7. Считаем оконную функцию и применяем словарь

```
SELECT
    dictGet('user_email', 'email', user_id) AS email,
    action,
    sum(expense) OVER (PARTITION BY action ORDER BY email DESC)
```

```
FROM user_action

Query id: b2d6d40d-cd86-4de8-a33b-faeb3a9ce35f

   ┌─email─────┬─action─┬─sum(expense)⋯email DESC)─┐
1. │ as@as.ru  │ drop   │                        3 │
2. │ as@as.ru  │ store  │                        2 │
3. │ ad@ad.ru  │ store  │                        4 │
   └───────────┴────────┴──────────────────────────┘
```

Заголовок